IGCSE 0607 Practice Questions — E2.8

Proportion

Total Questions: 26 — **Total Marks:** 157 Marks

Table of Contents & Quick Navigation

- **E2.8** Proportion
  → **Paper 2 (Non-Calculator)** *(12 Qs)* · **Paper 4 (Calculator)** *(14 Qs)*

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1 y is directly proportional to the square of x.
When $x = 3$, $y = 36$.

(a) Find an equation connecting y and x.

$y =$ ..... **[2]**

(b) Find the value of y when $x = 5$.

$y =$ ..... **[1]**

2 y is directly proportional to $(x + 2)$.
When $x = 3$, $y = 20$.

Find the value of y when $x = 7$.

$y =$ ..... **[3]**

3 y is directly proportional to the square of x.
When $x = 3$, $y = 36$.

(a) Find an equation connecting y and x.

$y =$ ..... **[2]**

(b) Find the possible values of x when $y = 64$.

$x =$ ..... **[2]**

4 y is directly proportional to $(x + 2)^2$.
When $x = 3$, $y = 75$.

(a) Find an equation for y in terms of x.

$y =$ ..... **[2]**

(b) Calculate the value of y when $x = 6$.

$y =$ ..... **[2]**

5 y is directly proportional to x.
When $x = 5$, $y = 15$.

(a) Find an equation connecting y and x.

..... **[2]**

(b) Work out the value of y when $x = 12$.

$y =$ ..... **[2]**

6 y is directly proportional to $(x + 2)^3$.
When $x = 1$, $y = 135$.

(a) Find an equation for y in terms of x.

$y =$ ..... **[2]**

(b) Find the value of y when $x = 0$.

$y =$ ..... **[1]**

(c) Find the value of x when $y = 320$.

$x =$ ..... **[2]**

7 The variable y is directly proportional to the square of x.
When $x = 3$, $y = 36$.

(a) Find an equation connecting y and x.

$y =$ ..... **[2]**

(b) Find the value of y when $x = 5$.

$y =$ ..... **[1]**

(c) Find the values of x when $y = 196$.

$x =$ ..... **[2]**

8 y is directly proportional to the cube of x.
When $x = 2$, $y = 24$.

(a) Find an equation connecting y and x.

$y =$ ..... **[2]**

(b) Work out the value of y when $x = 5$.

$y =$ ..... **[1]**

(c) Work out the value of x when $y = 81$.

$x =$ ..... **[2]**

9 The table shows some corresponding positive values of p and q.

p	2	4	10
q	100	25	4

(a) Determine the algebraic equation connecting q and p.

$q =$ ..... **[2]**

(b) Find the value of q when $p = 5$.

$q =$ ..... **[1]**

(c) Find the value of p when $q = 1$.

$p =$ ..... **[2]**

10 The resistance, R ohms, of a uniform cylindrical wire is directly proportional to its length, L metres, and inversely proportional to the square of its diameter, d millimetres.
When $L = 12$ and $d = 2$, $R = 9$.

(a) Find a formula for R in terms of L and d.

$R =$ ..... **[3]**

(b) A second wire made of the same material has length 40 metres and resistance 7.5 ohms.
Calculate the diameter, d, of this second wire.

$d =$ ..... mm **[2]**

(c) The length of a wire is increased by 25% and its diameter is halved.
Find the percentage increase in its resistance.

..... % **[2]**

11 P is directly proportional to the square root of Q.
Q is inversely proportional to R^3.
When $Q = 9$, $P = 12$.
When $R = 2$, $Q = 2$.

(a) Find an equation for P in terms of R.

$P =$ ..... **[5]**

(b) Calculate the value of P when $R = 4$.

$P =$ ..... **[2]**

12 A is directly proportional to the square of B.
B is inversely proportional to $(C + 2)$.
Assume that $B > 0$ and $(C + 2) > 0$.
When $B = 4$, $A = 80$.
When $C = 2$, $B = 3$.

(a) Find the value of A when $C = 4$.

$A =$ ..... **[4]**

(b) Find an expression for C in terms of A.

$C =$ ..... **[4]**

13 The table shows corresponding values of two positive variables p and q.

p	4	9	25	64
q	18	12	7.2	4.5

(a) State which variation model best fits the data in the table, writing your answer using the proportional symbol $\propto$.

..... **[2]**

(b) Find an equation connecting p and q.

$q =$ ..... **[2]**

(c) Calculate the value of p when $q = 2.4$.

$p =$ ..... **[2]**

14 y is inversely proportional to the square of x.
When $x = 4$, $y = 3$.

(a) Find an equation connecting y and x.

..... **[3]**

(b) Work out the possible values of x when $y = 0.75$.

$x =$ ..... **[3]**

15 P is inversely proportional to the square of Q.
When $Q = 4$, $P = 1.5$.

(a) Find an equation connecting P and Q.

$P =$ ..... **[3]**

(b) Find the value of P when $Q = 10$.

$P =$ ..... **[1]**

(c) Find the positive value of Q when $P = 24$.

$Q =$ ..... **[2]**

16 The quantity P is inversely proportional to the square root of Q.
When $Q = 9$, $P = 12$.

(a) (i) Find the value of P when $Q = 144$.

$P =$ ..... **[3]**

(ii) Find the value of Q when $P = 1.5$.

$Q =$ ..... **[2]**

(b) Calculate the percentage decrease in P when the value of Q is multiplied by 4.

..... % **[2]**

17 M is inversely proportional to the square root of N.
When $N = 16$, $M = 15$.

(a) Find an equation connecting M and N.

$M =$ ..... **[2]**

(b) Find the value of M when $N = 100$.

$M =$ ..... **[2]**

(c) Find the value of N when $M = 1.2$.

$N =$ ..... **[3]**

18 The table shows some values of two variables, x and y.
The relationship between x and y can be modelled by a variation equation.

x	2	4	6
y	36	9	4

(a) Use the data to identify the correct variation model and find an equation connecting y and x.

$y =$ ..... **[3]**

(b) Find the positive value of x when $y = 2.25$.

$x =$ ..... **[2]**

(c) When x is multiplied by 3, y is multiplied by a constant p.
Find the value of p.

$p =$ ..... **[2]**

19 p is inversely proportional to the square root of q.
When $q = 144$, $p = 5$.

(a) Find an equation connecting p and q.

..... **[2]**

(b) Find the value of q when $p = 15$.

$q =$ ..... **[2]**

(c) The value of q is increased by 300%.
Find the percentage decrease in p.

..... % **[3]**

20 P is inversely proportional to the square root of Q.
When $Q = 36$, $P = 4$.

(a) Find an equation for P in terms of Q.

$P =$ ..... **[2]**

(b) Calculate the value of P when $Q = 144$.

$P =$ ..... **[1]**

(c) Calculate the value of Q when $P = 1.5$.

$Q =$ ..... **[2]**

(d) The value of Q is increased by 300%.
Calculate the percentage decrease in the value of P.

..... % **[3]**

21 **(a)** Some corresponding values of x and y are given below.

$$x = 2,\ y = 75 \qquad x = 4,\ y = 18.75 \qquad x = 10,\ y = 3$$

Identify which of the following models best fits the data:

$$y \propto x \qquad y \propto x^2 \qquad y \propto \frac{1}{x} \qquad y \propto \frac{1}{x^2}$$

Hence, find the equation for y in terms of x.

$y =$ ..... **[3]**

(b) P is directly proportional to the cube of Q.
When Q is increased by 50%, work out the percentage increase in P.

..... % **[3]**

22 **(a)** y is directly proportional to the cube of x.

When x increases by 20%, find the percentage increase in y.

..... % **[3]**

(b) The table shows some corresponding values of v and A.

Determine the best variation model connecting A and v from the following list:

$$A \propto v \qquad A \propto v^2 \qquad A \propto \sqrt{v} \qquad A \propto \frac{1}{v}$$

You must show working to justify your decision.

v	2	4	10
A	12	48	300

..... **[3]**

23 **(a)** The table shows some values of x and y.

x	2	4	5	8
y	32	8	5.12	2

(i) Identify the variation model and write down the equation for y in terms of x.

$y =$ ..... **[3]**

(ii) Calculate the value of y when $x = 10$.

$y =$ ..... **[1]**

(b) A is directly proportional to the cube of B.
When B is increased by $x\%$, the value of A increases by 72.8%.
Find the value of x.

$x =$ ..... **[3]**

24 **(a)** y is inversely proportional to the square root of x.

When x is increased by 44%, find the percentage change in y.

..... % **[4]**

(b) P is directly proportional to the cube of Q.
Q is inversely proportional to R.
When $R = 2$, $P = 10$.

Find the value of P when $R = 4$.

$P =$ ..... **[3]**

25 The table shows some corresponding values of x and y.

x	2	4	8
y	100	25	6.25

(a) Determine the relationship between x and y and find an equation connecting them.

..... **[3]**

(b) z is directly proportional to x.
When $x = 4$, $z = 10$.

(i) Find an equation for x in terms of z.

$x =$ ..... **[2]**

(ii) Hence find an equation for y in terms of z.
Give your answer in its simplest form.

$y =$ ..... **[3]**

26 The table shows some corresponding values of x and y.

x	2	4	10
y	40	5	0.32

(a) Determine the proportional relationship between y and x and find an equation connecting them.

$y =$ ..... **[3]**

(b) Find the value of x when $y = 2560$.

$x =$ ..... **[2]**

(c) The variable x is inversely proportional to w.
When $w = 5$, $x = 2$.

Find an equation connecting y and w.
Give your answer in its simplest form.

$y =$ ..... **[4]**